JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURAJADA VIZIANAGARAM

SRI SIVANI COLLEGE OF ENGINEERING (AFFILIATED TO JNTU-GV)

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

★ OFFICIAL ACADEMIC SYLLABUS & BLUEPRINT ★

B.Tech EEE R23 Electrical Circuit Analysis (ECA) Syllabus Blueprint

Course Code:	R23-EEE-1102	**Academic Scheme:**	I B.Tech I Semester (R23 Regulation)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Unit I: Basic Circuit Concepts & Reduction Techniques

Circuit elements, Voltage and current sources, Kirchhoff's laws, Star-Delta transformations, Source shifting, Nodal and Mesh analysis with dependent sources.

Unit II: Single Phase AC Circuits & Phasor Diagrams

Sinusoidal alternating quantities, RMS and average values, Form factor, Peak factor, Concept of phasor, Complex power, Apparent, active and reactive power, Power factor improvement.

Unit III: Network Theorems applied to AC & DC Circuits

Superposition theorem, Thevenin's and Norton's theorems, Maximum power transfer theorem, Millman's theorem, Tellegen's theorem.

Unit IV: Three-Phase Balanced & Unbalanced Systems

Phase sequence, Star and Delta connections, Relation between line and phase voltages and currents in balanced systems, Measurement of 3-phase power by Two-Wattmeter method.

Unit V: Magnetic Circuits & Coupled Inductors

Faraday's laws of electromagnetic induction, Self and mutual inductances, Reluctance, MMF, Flux, B-H curve, Series and parallel magnetic circuits, Ideal transformer model.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Continuous Internal Evaluation	30 Marks
Semester Examination	70 Marks
Credits	3.0 (3-0-0)

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM